

THE MEMBRANE INTERFACE AS A MASTER REGULATOR OF CELL DIVISION ORIENTATION IN PLANTS

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ABSTRACT

Cellular membranes are dynamic functional interfaces that orchestrate communication between cells and their environment. Plasma membrane lipid composition finely tunes cellular functions in space and time, notably by regulating membrane trafficking and the cytoskeleton. However, how membrane lipids control cell division in tissues remains poorly understood. Using the plant model *Arabidopsis*, we combine quantitative imaging and computational modelling to show that the phosphoinositide PI(4,5)P₂ is tightly patterned at the cell plate during cytokinesis, forming opposing polarity domains that likely coordinate membrane trafficking and cytoskeleton regulation. In mutant impaired in the plant specific phosphoinositide phosphatase SAC9, misregulation of these lipid patterns leads to ectopic division sites, revealing a hidden mechanical landmark at tricellular junctions that attracts the new cell wall. Our findings identify membrane lipids as central hubs integrating biochemical and mechanical signals within the cell wall–membrane–cytoskeleton continuum, thereby controlling division orientation and tissue topology in plants.

website: <https://www.ens-lyon.fr/RDP/signalisation-cellulaire-et-endomembranes/>